However, unlike the electrical signal going into a speaker, the pattern of a radio wave doesn't directly correspond to the sound wave. Instead, the information is encoded onto a carrier wave via modulation of either its amplitude (AM) or frequency (FM). The pattern of the modulation is usually what corresponds to the audio signal. FM radio has lower noise and affords better quality than AM radio, however we even have digital radio which can transmit CD-like quality over FM radio.

AirPlay and DLNA

Back in the realm of the everyday individual's personal electronics, almost everything around the house is connected to the WiFi. If you own an Apple device that stores your media, and have an Apple TV connected to the HDMI IN port of your media playing system, you're using their proprietary AirPlay protocol to stream your media over the home network. While this is smart and convenient, you are limited to the Apple ecosystem. The most common and easiest to use non-Apple alternative is the DLNA (Digital Life Network Alliance) standard. Chances are, your home theatre will be DLNA compliant, and if so it will automatically show up as a playback device on your Android/PC under the "Play To" menu. The speed and bandwidth of WiFi allows you to enjoy high quality audio without having to plug in any cables.

Bluetooth

Whether it's listening to music in your portable speaker, or talking on the phone whilst retaining the mobility and freedom of your hands, the most ubiquitous wireless audio technology of today (at least on the road) is definitely Bluetooth! Named after Danish king Harald Bluetooth, this wireless communication technology was invented in the labs of Ericsson Mobile in Lund, Sweden. It was designed for short range data transfer and uses Ultra High Frequency (UHF) radio waves to communicate over the 2.4GHz - 2.485GHz band. Bluetooth networks are Personal Area Networks and therefore generally limited to a range of about 10 meters. Devices need to have a Bluetooth chip to communicate with each other, and the data transfer rate, range, and bandwidth are much lower than that of WiFi, but Bluetooth scores big on convenience and mobility, and doesn't need an intermediary router.